CHM 102 PREP QUESTION WITH COSMAS@EKSU 01

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1. What is the general formula of alkynes?
a) CnH2n
b) CnH2n+2
c) CnHn
d) CnH2n-2

2. Which hydrocarbon is known as the simplest alkene?
a) Ethene
b) Propene
c) Methane
d) Butene

3. What type of isomerism is exhibited by butane and isobutane?
a) Structural isomerism
b) Geometric isomerism
c) Optical isomerism
d) Conformational isomerism

4. The combustion of hydrocarbons produces primarily:
a) Water and carbon dioxide
b) Oxygen and nitrogen
c) Sulfur dioxide and water
d) Carbon monoxide and methane

5. Which hydrocarbon is commonly used as a fuel in cigarette lighters?
a) Butane
b) Propane
c) Ethylene
d) Methane

6. What is the IUPAC name of CH3-CH=CH2?
a) Propene
b) Butene
c) Ethene
d) Pentene

7. Aromatic hydrocarbons contain at least one:
a) Double bond
b) Triple bond
c) Benzene ring
d) Alkyne group

8. Which process is commonly used to convert alkenes to alkanes?
a) Hydrogenation
b) Halogenation
c) Oxidation
d) Isomerization

9. How many carbon atoms are there in the benzene ring?
a) 4
b) 5
c) 6
d) 7

10. What is the primary source of natural gas, a mixture of hydrocarbons?
a) Petroleum
b) Coal
c) Natural gas fields
d) Shale gas

11. Which hydrocarbon is a major component of natural gas?
a) Methane
b) Ethane
c) Propane
d) Butane

12. The addition of chlorine to an alkane is an example of:
a) Substitution reaction
b) Addition reaction
c) Elimination reaction
d) Oxidation reaction

13. What is the common name for C2H6?
a) Ethene
b) Ethyne
c) Ethane
d) Methane

14. Which hydrocarbon is the main component of gasoline?
a) Methane
b) Ethane
c) Propane
d) Octane

15. In alkanes, carbon atoms are bonded together by:
a) Single bonds
b) Double bonds
c) Triple bonds
d) Ionic bonds

16. What is the purpose of recrystallization in the isolation of organic compounds?
a) To remove impurities
b) To increase solubility
c) To enhance color
d) To accelerate reaction rates

17. Which technique is commonly used for separating organic compounds based on their boiling points?
a) Filtration
b) Distillation
c) Chromatography
d) Crystallization

18. What is the role of a desiccant in the drying process during organic compound isolation?
a) To add moisture
b) To remove water
c) To increase solubility
d) To enhance color

19. In column chromatography, separation is achieved based on differences in:
a) Density
b) Refractive index
c) Boiling point
d) Adsorption

20. What is the primary purpose of extraction in the isolation of organic compounds?
a) To increase yield
b) To remove impurities

c) To enhance color
d) To accelerate reaction rates

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1. b) CnH2n+2

2. a) Ethene

3. a) Structural isomerism

4. a) Water and carbon dioxide

5. a) Butane

6. b) Butene

7. c) Benzene ring

8. a) Hydrogenation

9. c) 6

10. c) Natural gas fields

11. a) Methane

12. b) Addition reaction

13. c) Ethane

14. d) Octane

15. a) Single bonds

16. a) To remove impurities

17. b) Distillation

18. b) To remove water

19. d) Adsorption

20. b) To remove impurities

21. In vacuum filtration, what is the function of applying reduced pressure?
a) Accelerate filtration
b) Increase temperature
c) Remove impurities
d) Enhance crystallization

22. Which solvent is often used for washing organic compounds during the purification process?
a) Water
b) Hexane
c) Ethanol
d) Acetone

23. What is the purpose of adding a drying agent during the isolation of organic compounds?
a) Increase solubility
b) Remove water
c) Enhance color
d) Accelerate reaction rates

24. Which method is suitable for separating a mixture of compounds with similar polarities?
a) Distillation
b) Filtration
c) Chromatography
d) Crystallization

25. Why is refluxing often employed in the purification of organic compounds?
a) To enhance color

b) To accelerate reaction rates
c) To remove impurities
d) To increase solubility

26. Who is often referred to as the "Father of Organic Chemistry"?
a) Dmitri Mendeleev
b) Robert Boyle
c) Friedrich Wöhler
d) Linus Pauling

27. The term "organic chemistry" originally referred to compounds derived from:
a) Minerals
b) Living organisms
c) Metals
d) Gases

28. What landmark discovery by Friedrich Wöhler challenged the belief in vitalism?
a) Synthesis of urea
b) Discovery of benzene
c) Isolation of methane
d) Formation of aldehydes

29. The concept of valence, vital to organic chemistry, was introduced by:
a) Gilbert Lewis
b) Linus Pauling
c) August Kekulé
d) Antoine Lavoisier

30. Who proposed the theory of resonance in organic molecules?
a) Robert Hooke
b) Linus Pauling
c) Friedrich Wöhler
d) Linus Pauling

31. The structure of benzene, with alternating single and double bonds, was suggested by:
a) Friedrich Wöhler
b) August Kekulé
c) Robert Boyle
d) Dmitri Mendeleev

32. The IUPAC (International Union of Pure and Applied Chemistry) system for naming organic compounds was established in:
a) 1850
b) 1900
c) 1920
d) 1950

33. Who shared the Nobel Prize in Chemistry in 1954 for his research on the nature of the chemical bond and its application to the elucidation of the structure of complex substances?
a) Robert Hooke
b) Linus Pauling
c) Gilbert Lewis
d) Antoine Lavoisier

34. The modern understanding of stereochemistry was advanced by the work of:
a) Robert Boyle
b) Louis Pasteur
c) Dmitri Mendeleev
d) Linus Pauling

35. The term "isomerism" in organic chemistry refers to compounds with the same molecular formula but different:
a) Melting points
b) Boiling points
c) Structures
d) Colors

36. Which of the following hydrocarbons contains only single bonds?
a) Alkene
b) Alkyne
c) Alkane
d) Aromatic hydrocarbon

37. The general formula for alkynes is:
a) CnH_2n
b) CnH_2n+2
c) CnHn
d) CnH_2n-2

38. What is the common name for hydrocarbons with at least one carbon-carbon double

bond?
a) Alkanes
b) Alkenes
c) Alkynes
d) Aliphatic hydrocarbons

39. Which type of hydrocarbon exhibits a “pi” bond?
a) Alkane
b) Alkene
c) Alkyne
d) Aromatic hydrocarbon

40. The process of adding hydrogen to unsaturated hydrocarbons is called:
a) Isomerization
b) Hydrogenation
c) Halogenation
d) Dehydrogenation

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1. **a) Accelerate filtration**
2. **a) Water**
3. **b) Remove water**
4. **c) Chromatography**
5. **c) To remove impurities**
6. **c) Friedrich Wöhler**
7. **b) Living organisms**
8. **a) Synthesis of urea**
9. **c) August Kekulé**
10. **b) Linus Pauling**
11. **b) August Kekulé**
12. **c) 1920**
13. **b) Linus Pauling**

14. **b) Louis Pasteur**
15. **c) Structures**
16. **c) Alkane**
17. **d) $CnH_2n–2$**
18. **b) Alkenes**
19. **b) Alkene**
20. **b) Hydrogenation**
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41. What is the IUPAC name for the simplest alkene?
a) Methene
b) Ethene
c) Propene
d) Butene

42. Aromatic hydrocarbons are characterized by the presence of:
a) Single bonds
b) Double bonds
c) Triple bonds
d) Benzene ring

43. The general formula for alkynes is:
a) CnH2n
b) CnH2n+2
c) CnHn
d) CnH2n-2

44. Which hydrocarbon is commonly used as a fuel in Bunsen burners?
a) Methane
b) Ethene
c) Propane
d) Butene

45. How many sigma bonds are present in a molecule of ethene (C_2H_4)?
a) 1
b) 2
c) 3
d) 4

46. The double bond in alkenes consists of a:
a) Sigma bond and a pi bond
b) Two sigma bonds

c) Two pi bonds
d) Sigma bond and a delta bond

47. What is the general formula for alkanes?
a) CnH2n
b) CnH2n+2
c) CnHn
d) CnH2n-2

48. Which hydrocarbon is found in natural gas?
a) Methane
b) Ethene
c) Propyne
d) Butane

49. What is the common name for C_2H_2?
a) Ethane
b) Ethene
c) Ethyne
d) Methane

50. How many pi bonds are present in a molecule of propene (C_3H_6)?
a) 1
b) 2
c) 3
d) 4

51. What is the correct IUPAC name for CH_3-CH_2-CH_2-CH_3?
a) Butane
b) Propane
c) Butene
d) Ethane

52. The IUPAC name for CH_3-$CH(CH_3)$-CH_2-CH_3 is:
a) 2-methylbutane
b) 3-methylbutane
c) 2-ethylbutane
d) 3-ethylbutane

53. How is a substituent indicated in an IUPAC name?
a) As a prefix

b) As a suffix
c) As an infix
d) As a superscript

54. The IUPAC name for CH_3-CH_2-CHO is:
a) Propanal
b) Butanal
c) Propionaldehyde
d) Butyraldehyde

55. Which of the following is the correct IUPAC name for C_6H_6?
a) Hexane
b) Benzene
c) Cyclohexane
d) Hexene

56. How are double bonds indicated in IUPAC nomenclature?
a) -ene
b) -ane
c) -one
d) -ol

57. What is the IUPAC name for CH_3-CH_2-CH(CH_3)-CH_2-CH_3?
a) 2-methylpentane
b) 3-methylpentane
c) 2-ethylpentane
d) 3-ethylpentane

58. The IUPAC name for CH_3-CH_2-OH is:
a) Ethanol
b) Methanol
c) Propanol
d) Butanol

59. What is the IUPAC name for CH_3-CH_2-COOH?
a) Acetic acid
b) Formic acid
c) Butyric acid
d) Propionic acid

60. In the IUPAC name CH_3-CH_2-CH_2-CH_2-CH_3, the root word is:

a) But
b) Pent
c) Hex
d) Hept

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41. b) Ethene

42. d) Benzene ring

43. d) CnH2n-2

44. c) Propane

45. b) 2

46. a) Sigma bond and a pi bond

47. b) CnH2n+2

48. a) Methane

49. c) Ethyne

50. b) 2

51. a) Butane

52. a) 2-methylbutane

53. a) As a prefix

54. a) Propanal

55. b) Benzene

56. a) -ene

57. a) 2-methylpentane

58. a) Ethanol

59. a) Acetic acid

60. b) Pent

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61. The IUPAC name for the compound CH3-CH2-CH(Cl)-CH3 is:
a) 2-chlorobutane
b) 3-chlorobutane
c) 2-chloropentane
d) 3-chloropentane

62. How are substituent positions indicated in the IUPAC name?
a) Alphabetical order
b) Numerical locants
c) Random order
d) Molecular weight

63. What is the IUPAC name for CH3-CHO?
a) Methanal
b) Ethanal
c) Propanal
d) Butanal

64. How are multiple substituents indicated in IUPAC nomenclature?
a) Alphabetical order
b) Numerical locants
c) Random order
d) Molecular weight

65. The IUPAC name for CH3-CH2-CH2-CH2-CH2-CH3 is:
a) Hexane
b) Pentane
c) Heptane
d) Octane

66. What type of analysis focuses on identifying the components or elements present in a sample?
a) Qualitative analysis
b) Quantitative analysis
c) Both
d) None

67. In qualitative analysis, the presence of a specific element is often confirmed by:
a) Measuring its concentration
b) Using a standard curve
c) Visual observations
d) Calculating molarity

68. Which technique is commonly used for qualitative analysis of organic compounds based on their mass-to-charge ratio?
a) Mass spectrometry
b) NMR spectroscopy
c) Infrared spectroscopy
d) UV-Vis spectroscopy

69. What is the primary goal of quantitative analysis?
a) Identifying components
b) Determining concentrations
c) Visual observations
d) Qualitative assessments

70. In quantitative analysis, calibration curves are often employed to relate:
a) Absorbance to concentration
b) Mass to volume
c) Color to identity
d) Density to temperature

71. Titration is a common technique used in quantitative analysis to determine:
a) Mass

b) Volume
c) Concentration
d) Color

72. Gravimetric analysis is a quantitative technique based on:
a) Volume measurements
b) Mass measurements
c) Spectral observations
d) Color changes

73. Which of the following is a qualitative method for determining the presence of specific ions in a solution?
a) Chromatography
b) Titration
c) Flame test
d) Mass spectrometry

74. What is the key difference between qualitative and quantitative analysis?
a) Instruments used
b) Purpose
c) Sample size
d) Time required

75. Which analytical technique is commonly used for determining the concentration of a colored compound in a solution?
a) UV-Vis spectroscopy
b) Infrared spectroscopy
c) Mass spectrometry
d) NMR spectroscopy

76. What does valence represent in the context of chemistry?
a) Atomic mass
b) Electron configuration
c) Bonding capacity
d) Mass number

77. How many valence electrons does an atom of chlorine have?
a) 1
b) 2
c) 7
d) 8

78. The valence of an element is often associated with the number of:
a) Neutrons
b) Electrons in the inner shell
c) Electrons in the outermost shell
d) Protons

79. The concept of valence force is related to the:
a) Strength of chemical bonds
b) Size of atoms
c) Electronegativity of elements
d) Atomic radius

80. How does the valence force influence the reactivity of an element?
a) Higher valence force increases reactivity
b) Lower valence force increases reactivity
c) Valence force is unrelated to reactivity
d) Valence force decreases reactivity

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61. a) 2-chlorobutane

62. b) Numerical locants

63. b) Ethanal

64. a) Alphabetical order

65. a) Hexane

66. a) Qualitative analysis

67. c) Visual observations

68. a) Mass spectrometry

69. b) Determining concentrations

70. a) Absorbance to concentration

71. c) Concentration

72. b) Mass measurements

73. c) Flame test

74. b) Purpose

75. a) UV-Vis spectroscopy

76. c) Bonding capacity

77. c) 7

78. c) Electrons in the outermost shell

79. a) Strength of chemical bonds

80. a) Higher valence force increases reactivity

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81. Which element has a valence electron configuration of ns^2np^6?
a) Oxygen
b) Nitrogen
c) Carbon
d) Neon

82. The valence force between atoms is responsible for:

a) Atomic size
b) Chemical bonding
c) Ionization energy
d) Electronegativity

83. What is the maximum number of valence electrons an atom can have in its outermost shell?
a) 2
b) 6
c) 8
d) 10

84. The valence electrons of an atom are involved in:
a) Core electron interactions
b) Chemical bonding
c) Nuclear reactions
d) Ionization processes

85. How is the valence force related to the stability of a chemical bond?
a) Higher valence force increases stability
b) Lower valence force increases stability
c) Valence force is unrelated to stability
d) Valence force decreases stability

86. The arrangement of atoms in a crystal lattice defines its:
a) Molecular weight
b) Crystal structure
c) Atomic radius
d) Covalent bonding

87. In a face-centered cubic (FCC) crystal structure, the atoms are located at the corners and the centers of:
a) Only the faces
b) Only the edges
c) Both faces and edges
d) The body diagonals

88. The repeating three-dimensional pattern in a crystal lattice is known as the:
a) Unit cell
b) Atomic radius
c) Ionic bond

d) Covalent network

89. Which type of solid has a well-defined geometric arrangement of atoms and strong covalent bonds?
a) Molecular solid
b) Metallic solid
c) Ionic solid
d) Network solid

90. The coordination number of an atom in a crystal lattice refers to:
a) The number of atoms it touches
b) Its atomic mass
c) Its electronegativity
d) Its valence electrons

91. The crystal lattice of sodium chloride (NaCl) is classified as:
a) Simple cubic
b) Body-centered cubic
c) Face-centered cubic
d) Hexagonal close-packed

92. Which type of bond is predominant in metallic solids?
a) Covalent bonds
b) Ionic bonds
c) Metallic bonds
d) Hydrogen bonds

93. The term "allotropy" refers to different:
a) Crystal structures of the same element
b) Elements in the same group
c) Types of solids
d) Bonding arrangements in a crystal lattice

94. The packing efficiency of a crystal lattice is the ratio of:
a) Empty space to occupied space
b) Atomic mass to molecular weight
c) Covalent bonds to metallic bonds
d) Ionic radius to atomic radius

95. The presence of delocalized electrons characterizes the bonding in:
a) Ionic solids

b) Molecular solids
c) Metallic solids
d) Network solids

96. Which element is found at the top of Group 18 (VIII A) in the periodic table?
a) Helium
b) Neon
c) Argon
d) Krypton

97. In the periodic table, elements in the same group share similar:
a) Atomic numbers
b) Atomic masses
c) Electron configurations
d) Ionic radii

98. The periodicity of chemical properties is best explained by the arrangement of elements based on their:
a) Mass number
b) Atomic radius
c) Electron configuration
d) Covalent bonding

99. What is the general trend in atomic radius across a period in the periodic table?
a) Increases
b) Decreases
c) Remains constant
d) No specific trend

100. Elements in the same period of the periodic table have the same number of:
a) Protons
b) Neutrons
c) Electrons
d) All of the above

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81. d) Neon

82. b) Chemical bonding

83. c) 8

84. b) Chemical bonding

85. b) Lower valence force increases stability

86. b) Crystal structure

87. c) Both faces and edges

88. a) Unit cell

89. d) Network solid

90. a) The number of atoms it touches

91. c) Face-centered cubic

92. c) Metallic bonds

93. a) Crystal structures of the same element

94. a) Empty space to occupied space

95. c) Metallic solids

96. a) Helium

97. c) Electron configurations

98. c) Electron configuration

99. b) Decreases

100. c) Electrons

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